

Refinery Turnaround and Inspection Planning Checklist

FREE refinery turnaround checklist PDF. Refinery turnaround and shutdown inspection planning covering mec...

90-120 minutes | Per Turnaround (Typically Every 4-6 Years) | Inspection Manager / Reliability Engineer

Standards: [API 510 - Pressure Vessel Inspection Code: In-Service Inspection](#), [API 570 - Piping Inspection Code](#), [API 653 - Tank Inspection, Repair, Altera](#)

Inspection Information

Process Unit Name *

Planned Turnaround Start Date *

____ / ____ / ____

Lead Inspector Name *

Planned Turnaround Duration (Days)

Risk-Based Inspection Scope Development

Review RBI-driven inspection scope.

1. RBI assessment current for all equipment in unit? * ☐ Yes ☐ No

API 580/581 Risk-Based Inspection (RBI) methodology assesses probability of failure (PoF) and consequence of failure (CoF) to optimize insp...

2. Equipment risk ranking matrix updated with most recent inspection data? * ☐ Yes ☐ No

3. Inspection intervals justified by RBI or code-maximum intervals met? * ☐ Yes ☐ No

4. All deferred inspection items from last turnaround included in current scope? * ☐ Yes ☐ No

5. Any new equipment since last T/A included in inspection scope? * ☐ Yes ☐ No

Pressure Vessel Inspection (API 510)',

Plan pressure vessel inspection scope.

1. All pressure vessels due for API 510 internal inspection listed in scope? * ☐ Yes ☐ No

2. Individual inspection plans developed for high-risk vessels? * ☐ Yes ☐ No

3. Corrosion monitoring location (CML) thickness measurements planned? * ☐ Yes ☐ No

4. NDE methods (UT, RT, PT, MT, AUT) selected for each vessel per damage m... ☐ Yes ☐ No

5. API 510 certified inspectors assigned to all vessels? * ☐ Yes ☐ No

Piping Inspection (API 570)

Plan piping circuit inspection.

1. All piping circuits identified and categorized by service class? * ☐ Yes ☐ No

2. Thickness measurement locations (TML) documented and accessible? * ☐ Yes ☐ No

3. Injection point inspection per API 570 Section 6.5.4 included? * ☐ Yes ☐ No

4. Dead legs and elbows inspection included? * ☐ Yes ☐ No

5. Soil-to-air interface inspection for buried piping? * ☐ Yes ☐ No ☐ N/A

Heat Exchanger Inspection

Plan heat exchanger inspection scope.

1. Heat exchanger bundle inspection (tubes, tubesheets, baffles) in scope? * ☐ Yes ☐ No
2. Eddy current or ultrasonic tube inspection for critical exchangers planned? * ☐ Yes ☐ No ☐ N/A
3. Chemical cleaning or hydro-blasting planned per fouling severity? * ☐ Yes ☐ No
4. Fitness-for-service evaluation planned for any degraded components? * ☐ Yes ☐ No ☐ N/A
5. Spare bundles or replacement tubes staged for potential repairs? * ☐ Yes ☐ No ☐ N/A

Storage Tank Inspection (API 653)

Plan aboveground storage tank inspection.

1. All ASTs due for API 653 internal inspection in scope? * ☐ Yes ☐ No ☐ N/A

API 653 requires that all aboveground storage tanks (AST) receive periodic inspections. Initial inspection within 10 years of service or per RBI ...

2. Tank bottom MFL or UT scan planned for internal inspection? * ☐ Yes ☐ No ☐ N/A
3. Shell weld examination and thickness measurement planned? * ☐ Yes ☐ No ☐ N/A
4. Tank bottom CP system inspection included? * ☐ Yes ☐ No ☐ N/A
5. API 653 certified inspector assigned? * ☐ Yes ☐ No ☐ N/A

Documentation and Close-Out

Plan inspection documentation and records.

1. Inspection records database current and accessible to inspection team? * ☐ Yes ☐ No
2. Inspection report templates prepared for all equipment types? * ☐ Yes ☐ No
3. Process for fitness-for-service evaluations during T/A documented? * ☐ Yes ☐ No
4. Repair procedures and contractor QC plan approved? * ☐ Yes ☐ No
5. Return-to-service requirements and final pressure test requirements docume... ☐ Yes ☐ No
6. PSM mechanical integrity records update plan for T/A findings? * ☐ Yes ☐ No

Summary & Sign-Off

Turnaround Inspection Readiness *

Scope Gaps and Actions Required

Scope Finalization Date

___ / ___ / ___

Lead Inspector Signature *
